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DATA 5100
October 21, 2025

Education Project: Modeling Relationships Between ACT Scores and Socio-Economic Factors

The objective of this project is to analyze the relationships between various socio-economic factors, and their effect on the average ACT score at a given school. Two datasets were provided, one from EdGap and another from the Common Core of Data (CCD). EdGap data contains our outcome variable, ACT score, as well as information about nearly 8,000 High Schools across 20 states. CCD data provides additional school characteristics of over 100,000 schools in the United States. Other datasets were joined to capture additional predictors such as the number of arrests and referrals to law enforcement per school, the student-teacher ratio, and the number of higher education institutions by zip code. Schools in states that mandate standardize testing are flagged. After extensive pre-processing and data cleansing, multiple multivariate linear regression models were compared. Ultimately, the percent of students receiving free or reduced lunch proved the most statistically significant and practical predictor, as it accounts for the majority of the explanatory power of each demonstrated model.

Introduction

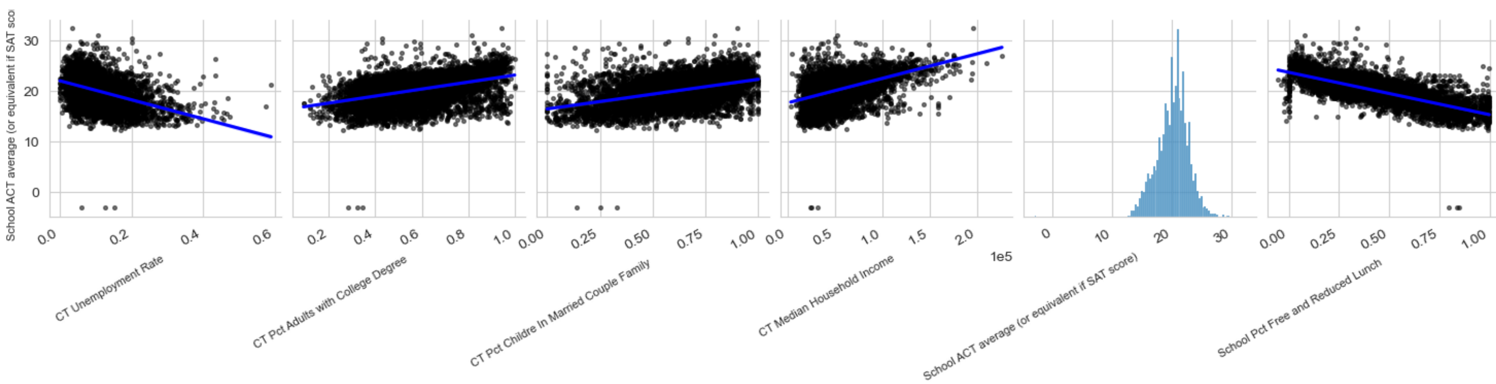
This project examines how some socio-economic factors, school characteristics, and local context influence ACT performance. The main dataset comes from EdGap, a resource designed to highlight disparities in college-readiness among schools; with the distinct mission of illustrating the 'achievement gap' as an 'opportunity gap'. EdGap provides two levels of data: School level data, such as Average ACT score and percent of students receiving free or reduced lunch and census tract data such as household income, unemployment, adult educational attainment, and family structure. ACT scores include scaled SAT scores. Detailed location data and information about each school from the Common Core of Data (CCD) is also provided. Additional data were sourced to create a more robust dataset. The number of arrests and/or referrals to law enforcement by school is found via the Department of Education's Civil Rights Data Collection (CRDC), with the hypothesis that the more referrals and arrests lead to lower ACT scores. This data was pulled for the 2017-2018 school year. For the same schoolyear, additional CCD data is obtained to gather student counts, teacher counts, student teacher ratios, a description of the area, and whether the school was a Magnet or Shared Time school. Two other data sources include the count of higher education institutions by zip code from the National Center for Education Statistics and a state-level flag was created to indicate whether college entrance exams such as the ACT or SAT are mandatory, provided, or optional. The former tests the hypothesis that schools near a college or university may have slightly higher ACT scores, these areas are more likely to provide greater academic exposure. The latter accounts for higher scores from testing-optional schools where students who opt in are biased towards college readiness. From this more robust data we are able to further the analysis brought forth by EdGap, and identify what socio-economic factors contribute most to college readiness benchmarks, in this case ACT scores.

Background

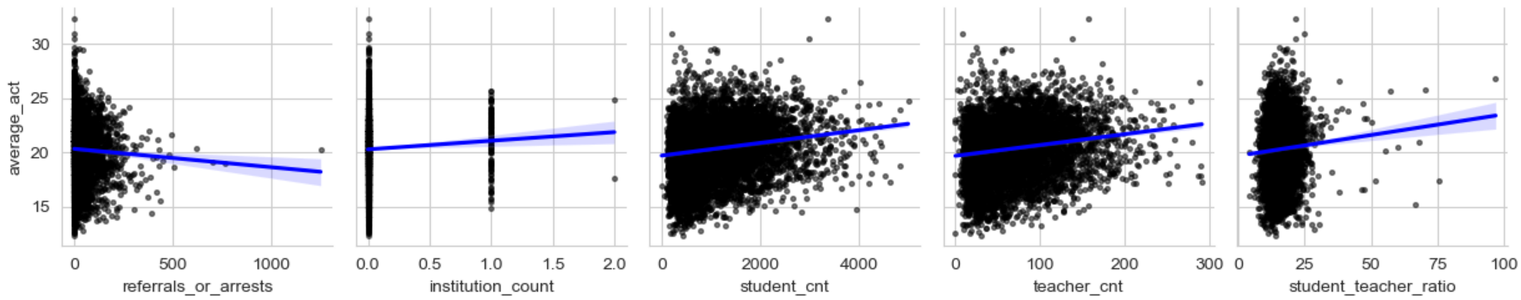
Understanding how schools and their locations contribute to educational attainment is an important discussion for various stakeholders. The 'School to Prison Pipeline' is a well-established phenomenon, and multiple studies have been dedicated to understanding what changes in schools lead to better served students. Additional data is added to the EdGap data for further analysis on these points. First, student/teacher ratio which is often used as a measure of how attentive a classroom can be. It is assumed this will be an indicator of improved ACT scores. Second, the number of students arrested or referred to law enforcement signals a challenging learning environment, and may be an indicator of lower ACT scores.

Methodology

Once all the data is imported, extensive preprocessing is required. Data from EdGap and the CCD School Information is relatively clean; columns are renamed and re-formatted for clarity and ease of use. Initial relationships within the EdGap data were verified visually using a pair plot, regression line, and checking individual variable distributions. It was determined that some multicollinearity existed between predictors, but not enough to cast doubt. A clear linear relationship can be observed between ACT scores and all of the predictive variables, warranting a regression analysis to quantify them fully.



The preprocessing of these data took longer than any other part of the analysis. Ensuring correct data types and precision for merging, especially where the school IDs were concerned, was a significant challenge. The supplemental CCD data contains numerous non-numeric symbols such as dashes (-), daggers (†) and double daggers (‡) which requires creative replacement. These data are located using their index positions and replaced with NaN in Pandas, ensuring consistent handling of missing values. Column names are standardized across all datasets and categorical data transformed. The variable "area_description" contains compound information, so the data are divided into distinct fields "area_type" and "area_size". Additionally, missing data in these datasets are set as NaN rather than imputed. Once the data is prepared and joined together, a secondary visual assessment of our relationship to ACT score is performed. These variables show less indication of direct relationships, but may produce value as additional predictors in a linear regression model:



For the final analysis, a robust multivariate linear regression model is selected as the method to quantify the relationships between our predictive variables and ACT scores. All variables are included in the robust model then backwards selection is applied to eliminate unnecessary predictors until the remaining data are relevant, statistically significant, and have sufficient explanatory power and fit. Using the statsmodels package, an Ordinary Least Squares model was constructed with the following formula:

```
multivariate_model = smf.ols(
    formula='average_act ~ rate_unemployment + percent_college + percent_lunch + student_cnt + teacher_cnt + student_teacher_ratio + institution_count + referrals_or_arrests + C(charter) + C(magnet_sch) + C(shared_time_sch) + C(testing) +  C(area_type) + C(area_size) + any_institutions + any_ref_or_arr',
    data=df_final).fit()
```

This model included 26 degrees of freedom, and many insignificant coefficients. Using a design matrix and backward selection, all coefficients with a p-value of 0.05 or greater were eliminated, and the model was rerun. Additional variables were backed out and re-run two more times before settling on 14 statistically significant variables to use in a robust model. As one final comparison, a highly reduced model was computed using only three EdGap predictors: rate_unemployment, percent_college, and percent_lunch. A single variate model was also assessed exploring the direct effect of percent_lunch on average_act.

Computational Results

According to the correlation matrix of all our numeric data, the strongest relationship to ACT scores is a negative correlation of 0.78 with the percent of students on free or reduced lunch. The three datapoints from EdGap have moderate to strong correlations to ACT Score. None of the data sourced independently produces a meaningful correlation to ACT Score; the highest being a positive correlation to the overall student count, of 0.18. Applying the categorical labels to the numeric data did not appear to tease out any additional meaning, when viewing pair plots.

The results of our linear regression modeling show that our highly reduced model of three EdGap variables can explain 96% of the variance captured by the robust model, with R-squared of 0.629 for the reduced and 0.651 for the robust. The reduced model accomplishes this using only 3 predictors instead of 14. All variables are statistically significant ($p < 0.05$) in both models, however the F-statistic for the reduced model is higher than that of the robust (4100 compared to 1030). Furthermore, the robust model presented with a multicollinearity warnings whereas the reduced model did not.

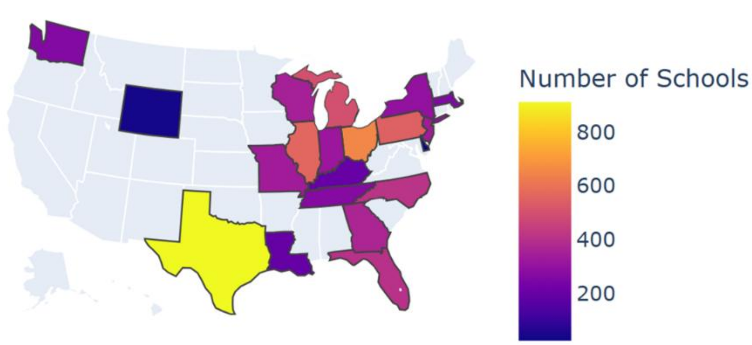
```
                            OLS Regression Results
==============================================================================
Dep. Variable:            average_act   R-squared:                       0.629
Model:                            OLS   Adj. R-squared:                  0.629
Method:                 Least Squares   F-statistic:                     4100.
Date:                Mon, 20 Oct 2025   Prob (F-statistic):               0.00
Time:                        20:56:26   Log-Likelihood:                -13380.
No. Observations:                7251   AIC:                         2.677e+04
Df Residuals:                    7247   BIC:                         2.680e+04
Df Model:                           3
Covariance Type:            nonrobust
=====================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
Intercept          22.6633      0.102    221.231      0.000      22.462      22.864
rate_unemployment  -2.2121      0.373     -5.923      0.000      -2.944      -1.480
percent_college     1.6881      0.127     13.308      0.000       1.439       1.937
percent_lunch      -7.6078      0.092    -82.404      0.000      -7.789      -7.427
==============================================================================
Omnibus:                      870.897   Durbin-Watson:                   1.490
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             3110.638
Skew:                           0.584   Prob(JB):                         0.00
Kurtosis:                       5.989   Cond. No.                         25.9
==============================================================================
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            average_act   R-squared:                       0.651
Model:                            OLS   Adj. R-squared:                  0.650
Method:                 Least Squares   F-statistic:                     1030.
Date:                Mon, 20 Oct 2025   Prob (F-statistic):               0.00
Time:                        20:40:36   Log-Likelihood:                -13055.
No. Observations:                7209   AIC:                         2.614e+04
Df Residuals:                    7195   BIC:                         2.623e+04
Df Model:                          13
Covariance Type:            nonrobust
=============================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------------
Intercept                  17.5739      0.147    119.146      0.000      17.285      17.863
C(shared_time_sch)[T.No]    0.2768      0.043      6.376      0.000       0.192       0.362
C(shared_time_sch)[T.Yes]  -0.6845      0.195     -3.511      0.000      -1.067      -0.302
C(testing)[T.Optional]      0.9270      0.054     17.202      0.000       0.821       1.033
C(testing)[T.Provided]      1.1582      0.059     19.604      0.000       1.042       1.274
C(area_type)[T.Rural]       3.3775      0.040     84.750      0.000       3.299       3.456
C(area_type)[T.Town]        3.5156      0.048     73.929      0.000       3.422       3.609
C(area_size)[T.Large]       3.2919      0.046     72.195      0.000       3.202       3.381
C(area_size)[T.Mid-size]    3.5439      0.068     52.240      0.000       3.411       3.677
C(area_size)[T.Small]       3.8450      0.066     58.088      0.000       3.715       3.975
any_ref_or_arr[T.False]     0.5335      0.097      5.493      0.000       0.343       0.724
any_ref_or_arr[T.True]      0.3552      0.094      3.787      0.000       0.171       0.539
rate_unemployment          -1.7367      0.373     -4.658      0.000      -2.468      -1.006
percent_college             1.9133      0.138     13.831      0.000       1.642       2.185
percent_lunch              -7.5695      0.099    -76.664      0.000      -7.763      -7.376
==============================================================================
Omnibus:                     1005.743   Durbin-Watson:                   1.569
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             4373.301
Skew:                           0.624   Prob(JB):                         0.00
Kurtosis:                       6.606   Cond. No.                     8.35e+15
==============================================================================
```

Comparison of models: reduced (left) vs. robust (right)

Discussion

Before interpreting the analysis, it is important to note some of the limitations of the data: The final dataset includes one year of data for 7,251 High Schools in 20 of 50 U.S. states. The West Coast and Midwest are severely underrepresented, as the vast majority of the states are in the



testing	average_act
Mandatory	19.808473
Optional	20.585132
Provided	20.104354

Great Lakes region. Furthermore, nearly 30% of the data comes from three states alone: Texas, Ohio and Illinois, where neither the SAT nor the ACT are mandatory. Texas is one of the six States in this sample that provides access to testing free of charge, but does not require students take them. These are important differences based on the population of students taking the tests, and a crucial source of bias in our sampling.

Given these data limitations, and despite our model significance, the results should be interpreted with caution. Because the reduced model has 96% of the explanatory power and uses 78% fewer predictors, it is the favorable linear model. Using an ANOVA comparison between the two linear models indicates the robust model has a statistically significant improvement over the reduced. Though the complexity cost of 11 additional variables for a 2% improvement in explanatory power (as indicated by the R-Squared) does not offer practical justification for utilizing the more complex model.

Conclusion

After significant pre-processing, cleaning, joining, and analysis, the most influential variable for predicting ACT scores in our sample comes from the percent of the student body receiving free or reduced lunch. This variable is specific to each school (rather than the surrounding area), and is a more direct measure of the school's economic need. Since families must qualify for free or reduced lunch via federal programs, it better normalizes and accounts for local cost-of-living differences between regions better compared to median income alone. While many of the variables introduced from sources outside of EdGap had some significant relationship, they failed to move the score in a meaningful way. The only variables in the more complex model that caused a significant shift in ACT scores were related to area type and size; however, nearly every type or size category contributed roughly the same increase of about 3 points, making these variables largely uninformative. In summary, while many of the data points included in the EdGap data and sourced elsewhere provide evidence of statistically significant predictive relationships to a school's average ACT score, the percent of students receiving free or reduced lunch proved to be the most meaningful. By omitting 11 of the 14 predictors used in the robust multivariate OLS model we still retain 96% of the explanatory power. The vast majority of the explanatory power in both these models comes from the 'percent_lunch' variable. In fact, the R-Squared of a single variate model that uses only the free and reduced lunch variable to predict average ACT score has an R-Squared of 0.616. The marginal benefit of adding each additional variable is nil, as we achieve an R-squared of 0.629 adding two more variables and 0.651 with a total of 14. Thus, we conclude that given the data available, a reduced 3 variable model strikes the right balance of model simplicity, explainability, accuracy, and goodness of fit.